

Doc. 265 – Formalism Correspondence: FFGFT and Blume-Maleev

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Purpose

This document provides a worked example for the structural comparison of physical formalisms, developed for the IPI group's model-ordering work. The example uses two independently derived mathematical descriptions of the same experimental system – the ferrotoroidic quaternary memory crystal $\text{LiNi}_{0.8}\text{Fe}_{0.2}\text{PO}_4$ (Qureshi et al. 2026) – to illustrate that "correspondence between formalisms" is not a single category but a stratified relation with at least three distinct levels.

The two descriptions are:

- **Blume-Maleev polarization matrix formalism:** the standard condensed-matter description, complete and quantitatively accurate, developed entirely within established experimental physics.
- **FFGFT (T^4 -torus geometry):** the geometric field theory framework, applied to the same system from its own first principles, without fitting or borrowing from the condensed-matter description.

Neither framework was designed with the other in mind. The correspondences that emerge are therefore structurally informative – they reveal what mathematical content is shared versus what is framework-specific.

Table 1: Classification of correspondence types between the Blume-Maleev and FFGFT descriptions of the ferrotoroidic system.

Level	Definition	Diagnostic criterion
1. Near-equivalence	Both formalisms express the same mathematical object, derived independently from different starting points.	The abstract structure (group, operator, invariant) is identical; only the derivation path and physical interpretation differ.
2. Structural overlap	The same qualitative structure appears in both, but the quantitative content is not shared.	One formalism can be mapped onto the other for the structural skeleton, but not for numerical predictions.
3. Complementary	Each formalism explains something the other cannot.	Neither is a subset or extension of the other; the union covers more than either alone.

.1 The Three Levels of Correspondence

.2 The comparison operator – made explicit

The three-level classification in Table 1 assigns each correspondence to a level. Up to this point, however, it remained *implicit by which criteria* this assignment is made – which operator decides whether a phenomenon falls on Level 1, 2 or 3. This section exposes that operator, so that it becomes inspectable and contestable rather than resting on intuition.

The six comparison axes

Two formalisms are compared along six independent axes. On each axis the content is either *shared* (both formalisms deliver the same) or *not shared*:

- 1. Shared invariants:** Do both produce the same conserved quantities / topological invariants? (e.g. the conserved winding / domain number)
- 2. Shared operators:** Does the same abstract operator structure appear? (e.g. the same group $\mathbb{Z}_2 \times \mathbb{Z}_2$, the same transformation rule)
- 3. Shared mathematical structures:** Do both share the qualitative scaffold – antisymmetric coupling, projection factor – even if the numerical values differ?
- 4. Shared predictions:** Do both deliver the same *quantitative* value for the same experiment?
- 5. Shared physical mechanisms:** Do both attribute the phenomenon to the *same* physical cause?
- 6. Shared hardware implications:** Do the same consequences for real devices / measurements follow from both?

How the axes define the levels

The level assignment is not intuition but reads directly off the pattern of shared axes: The assignment criterion is thus made explicit:

Table 2: The comparison operator: which axes are shared at which level. ✓ = shared, × = not shared, ~ = shared only qualitatively.

Comparison axis	Level 1 Near-equiv.	Level 2 Overlap	Level 3 Complement.
1. Invariants	✓	~	×
2. Operators	✓	~	×
3. Mathematical structure	✓	✓	×
4. Predictions (quantitative)	×	×	×
5. Physical mechanisms	×	×	×
6. Hardware implications	×	×	×

- **Level 1 (near-equivalence):** Axes 1–3 fully shared – identical invariant, identical operator, identical structure – with different derivation paths. Axes 4–6 remain separate. (Examples: $\mathbb{Z}_2 \times \mathbb{Z}_2$, $\mathcal{P}'' = 0$, topological charge.)
- **Level 2 (structural overlap):** Axis 3 shared (the same qualitative scaffold), axes 1–2 only partially, axes 4–6 not. (Examples: antisymmetric exchange, decompactification projection.)
- **Level 3 (complementary):** None of axes 1–3 shared; each formalism covers a domain the other cannot reach. (Examples: geometric necessity of $\mathbb{Z}_2 \times \mathbb{Z}_2$ in FFGFT; material-specific transition temperatures in Blume-Maleev.)

Notably, axes 4–6 (quantitative predictions, physical mechanisms, hardware) are shared at *no* level in this comparison. This is not a defect of the classification but its main finding: Blume-Maleev and FFGFT share mathematical *structure* (axes 1–3) but not physical *substance* (axes 4–6). This is precisely what distinguishes this correspondence from a genuine equivalence (see the section on the Heisenberg/Schrödinger analogy).

Residuals: limits of the operator

The operator made explicit here is itself not fully mature. Three residuals remain open and are declared here, not hidden:

- **Axis completeness:** Are six axes sufficient, or are dimensions missing (e.g. shared boundary conditions, shared symmetry-breaking patterns)? The list is a working state, not a proven basis.
- **Axis weighting:** All six axes are weighted equally here. Whether a shared invariant (axis 1) should weigh more than a shared structure (axis 3) is undecided.
- **Axis independence:** The axes are treated as independent but may not be – a shared operator structure (axis 2) often implies a shared invariant (axis 1). The correlation structure of the axes is not analysed.

These residuals do not change the level assignment in the present case (the pattern in Table 2 is robust), but they mark where the operator itself requires further examination. The value of making it explicit lies precisely here: a previously implicit but load-bearing criterion becomes an inspectable object – inspectable rather than persuasive.

.3 Level 1 – Near-Equivalences

The $\mathbb{Z}_2 \times \mathbb{Z}_2$ domain structure

In Blume-Maleev: The four magnetic domains are related by two symmetry operations of the crystal: time-inversion $1'$ (maps $a \leftrightarrow b$, the ferrotoroidic flip) and the 2-fold screw axis $2_1 \parallel y$ (maps $' \leftrightarrow ''$, the orientation flip). Together these generate a $\mathbb{Z}_2 \times \mathbb{Z}_2$ group structure over the four domains. This is derived from the crystal symmetry group $P1121/a'$ and the magnetic space group analysis.

In FFGFT: The T^4 torus has two independent binary degrees of freedom for closed winding loops: the temporal winding direction ($\pm \tilde{T}$, i.e. whether the loop winds clockwise or anti-clockwise in the compact time direction) and the spatial winding parity ($\pm c$, the two possible orientations of the winding plane in the observable 3D subspace). These generate the same $\mathbb{Z}_2 \times \mathbb{Z}_2$ group structure over the four winding classes.

Assessment: The same abstract group $\mathbb{Z}_2 \times \mathbb{Z}_2$ is derived from two completely different starting points – crystal symmetry on one side, T^4 compactification geometry on the other. This is a near-equivalence: the mathematical object is the same, the derivation paths are independent.

What it does not mean: This does not imply that the crystal lattice is a T^4 torus, or that the FFGFT parameters describe the specific crystal. The shared group structure is a mathematical fact about how binary symmetries combine – it does not identify the underlying physical mechanism.

The $\mathcal{P}'' = 0$ condition

In Blume-Maleev: The general polarization transformation is $\mathbf{P}_f = \mathcal{P}' \mathbf{P}_i + \mathcal{P}''$. For $\text{LiNi}_{0.8}\text{Fe}_{0.2}\text{PO}_4$, $\mathcal{P}'' = 0$ because the magnetic structure factors are purely imaginary: $\text{Re}(NM_{\perp i}^*) = 0$ and $\text{Im}(M_{\perp y} M_{\perp z}^*) = 0$. This follows from the antiferromagnetic head-to-tail configuration (moments related by inversion are antiparallel).

In FFGFT: The recursion operator $\mathcal{R}$ (Doc. 203) on the T^4 mode space decomposes into a rotational part (coherent mode transformation, winding number preserved) and a creation part (spontaneous mode generation at T^4 boundary nodes, Doc. 204). In the static T^4 ground state – no expansion, no horizon dynamics – the creation part vanishes: $\mathcal{P}'' = 0$.

Assessment: Both derive the same vanishing of the creation/annihilation term from a “no spontaneous generation” condition – in the crystal from the antiferromagnetic symmetry, in FFGFT from the static geometry. The mathematical structure is the same; the physical mechanism is different but yields the same formal result.

Topological charge conservation

In the paper: The memory is non-volatile – domain populations remain stable at $E = H = 0$ after field-cooling. The paper identifies this as a consequence of the energy barrier separating toroidal domains: switching requires applying a conjugate field $E \times H$ above a threshold.

In FFGFT: A mode with non-zero winding number n cannot decay to $n = 0$ without crossing the minimal energy of the T^4 boundary condition (Doc. 204, Fractal Boundary Condition). The winding number is a topological invariant – it is quantized and conserved unless a field large enough to overcome the topological barrier is applied.

Assessment: Both describe the same mathematical phenomenon – a discrete topological invariant protected by an energy barrier – though one frames it in terms of magnetic

anisotropy and the other in terms of T^4 compactification geometry. Near-equivalence at the level of the conservation law; different mechanisms at the level of the physical origin.

.4 Level 2 – Structural Overlap Without Equivalence

Antisymmetric exchange and the non-closure theorem

In Blume-Maleev: The Dzyaloshinskii-Moriya (DM) interaction is $\mathcal{H}^{\text{DM}} = \mathbf{D} \cdot (\mathbf{S}_1 \times \mathbf{S}_2)$, odd under spin exchange (antisymmetric), with $\mathbf{D}_{34} = -\mathbf{D}_{12}$ from inversion symmetry.

In FFGFT: The Non-Closure Theorem (Doc. 193) states that cross-products of mode labels from *different* winding classes are topologically inadmissible – they would generate hybrid modes that violate the closed winding structure of T^4 . The antisymmetric cross-product $S_i \times S_j$ maps structurally onto this: it couples two spin components (two different winding directions) in a way that is antisymmetric under interchange.

Assessment: The antisymmetric cross-product structure is shared. But the DM vector components D^b and D^c are material-specific quantities ($|D^b| < |D^c|$, at 35° from the c -axis) that have no FFGFT counterpart. FFGFT can say that the antisymmetric structure is topologically constrained; it cannot predict the specific DM vector orientation or magnitude for this crystal.

Decompactification projection

In Blume-Maleev: The factors p_{fx}/p_{ix} in the general polarization matrix (Eq. 9) measure the efficiency with which the measurement apparatus resolves the polarization state – how much of the abstract mathematical object is made observable by the specific experimental setup.

In FFGFT: The decompactification projection (Layer 1 to Layer 2, Doc. 001) measures the fraction of the unit-free geometric content that becomes observable in the SI-projected description. The ratio $T_{\text{CMB}}/E_\xi = (16/9)\xi^2$ is a dimensionless Layer-1 prediction; the absolute Kelvin value requires embedding in SI.

Assessment: The structure – a projection factor that mediates between the abstract mathematical content and the observable quantity – is the same in both. But the specific projection factors p_{fi} are apparatus-dependent, material-dependent numbers in the paper; in FFGFT the projection is governed by $\xi = 4/30000$ and the SI embedding. No quantitative mapping between the two is established.

.5 Level 3 – Complementary (Non-Overlapping)

What FFGFT explains that Blume-Maleev does not

The Blume-Maleev formalism derives the four-domain structure from the observed crystal symmetry. It takes the symmetry group as given and works out the consequences. It does not explain *why* the relevant symmetry group is $\mathbb{Z}_2 \times \mathbb{Z}_2$ rather than some other group – that is determined by the specific material and is taken from experiment.

FFGFT provides a geometric explanation: $\mathbb{Z}_2 \times \mathbb{Z}_2$ is the *necessary* structure for a compact T^4 torus with one time-like and one space-like compact direction contributing to the observable domain classification. The four-fold structure is not contingent on the material; it is a consequence of the topology.

Similarly, the non-volatility of the memory is an empirical finding in the paper. In FFGFT it is a structural necessity: topological charge conservation is a consequence of the T^4 boundary condition, not something that needs to be verified separately for each material.

What Blume-Maleev provides that FFGFT does not

The paper provides what FFGFT cannot: quantitative predictions for the specific crystal. The transition temperatures $T_2 = 25$ K and $T_4 = 21$ K, the domain population ratios (up to 90% majority domain under optimal field-cooling), the DM vector orientation (35° from c -axis), the polarization tensor elements P_{xy}/P_{yx} and P_{xz}/P_{zx} in Table 1 – all of these follow from the material parameters of $\text{LiNi}_{0.8}\text{Fe}_{0.2}\text{PO}_4$ and are outside FFGFT's scope. FFGFT has nothing to say about why the Ni/Fe ratio 0.8/0.2 produces four domains rather than two, or why the transition temperature is 21 K rather than 210 K.

.6 The Analogy: Why Not Heisenberg vs. Schrödinger?

A natural first analogy for two formalisms describing the same physics is Heisenberg's matrix mechanics vs. Schrödinger's wave mechanics (1925–26), which were subsequently proved to be fully equivalent (von Neumann 1932). But this analogy is too strong for the present case.

Heisenberg and Schrödinger were describing the *same physical objects* (atoms, electrons) at the *same scale* with the *same predictive scope*. Their equivalence meant that every prediction of one was also a prediction of the other.

The Blume-Maleev / FFGFT correspondence is weaker: the two formalisms share mathematical structure at Levels 1 and 2, but operate at different scales (angstrom lattice vs. sub-Planck T^4), with different predictive scope (material-specific quantitative vs. topological structural), and different derivation bases (experimental symmetry analysis vs. geometric first principles). There is no reason to expect full equivalence, and the Level 3 analysis confirms it does not hold.

A more precise analogy is the relation between:

- A symmetry-group classification (e.g. the Wigner classification of particles by irreducible representations of the Poincaré group): gives the topology, the quantum numbers, the conservation laws.
- A Lagrangian field theory for a specific particle (e.g. QED for the electron): gives all quantitative predictions for that specific particle.

The group classification is not a subset of QED, and QED is not a subset of the Wigner classification. They are complementary: the group gives the structural necessity (why the electron exists and has spin 1/2); the field theory gives the dynamics (how it interacts). Neither alone is sufficient.

Blume-Maleev corresponds to the field theory (quantitative, material-specific, dynamically complete). FFGFT corresponds to the group classification (structural necessity, scale-independent, topologically explanatory). The shared content at Levels 1 and 2 is the intersection – the topological structure that both descriptions must contain because it is a property of the physical system independent of the formalism used to describe it.

.7 Implications for Model Comparison

The three-level stratification has practical consequences for comparing frameworks in the IPI group:

1. **Level 1 correspondences are the most informative:** when two independently derived formalisms produce the same mathematical object (same group, same invariant, same conservation law), this is evidence that the object is a real structural feature of the physical system – not an artefact of the formalism. The $\mathbb{Z}_2 \times \mathbb{Z}_2$ structure appearing in both crystal symmetry analysis and T^4 geometry suggests that this group is a genuine feature of toroidal topology, not just a convenient description.
2. **Level 2 correspondences define the shared vocabulary:** the structural overlap tells us which concepts transfer between frameworks (antisymmetric coupling, projection factors) and which do not. This is useful for translation without overclaiming equivalence.
3. **Level 3 defines the genuine boundary:** each framework's exclusive content marks where the other cannot substitute for it. Recognizing this boundary prevents both overclaiming ("my framework explains everything the other does") and underclaiming ("the frameworks have nothing to do with each other").
4. **Full equivalence is the special case, not the default:** the Heisenberg/Schrödinger equivalence was a remarkable result, not a template. Most frameworks that share structural content are related by partial overlap or complementarity, not equivalence.

Reference. The ferrotoroidic experiment used as the test case: N. Qureshi et al., *Toroidicity as a route towards non-volatile quaternary memory in antiferromagnets*, *Nature Communications* **17**, 4033 (2026). DOI: 10.1038/s41467-026-70767-8. The FFGFT interpretation of all formulas in that paper is in Doc. 264 of this series.